HDECGCN: A Heterogeneous Dual Enhanced Network Based on Hybrid CNNs Joint Multiscale Dynamic GCNs for Hyperspectral Image Classification

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***Abstract*— A hyperspectral image (HSI) classification algorithm that combines graph convolutional networks (GCNs) and convolutional neural networks (CNNs) aims to generate complementary spatial-spectral joint information at the superpixel and pixel levels. However, the CNN part is typically a single 2-D or 3-D network that cannot fully capture the middle or long-range spatial relationships between pixels. Additionally, the GCNs part is commonly under-segmented in the superpixel segmentation process and does not consider the weight between neighboring superpixels when calculating the adjacency matrix. Therefore, this article proposes a multiscale dynamic tuning parameter, where the dual superpixel segmentation GCN strategy joins the enhanced hybrid 3-D–2-D CNN framework to enhance the superpixel and pixel complementary nature. The hybrid enhanced CNN branch uses the groupable convolutions with a mixed spectral stacking and residual nonlocal block at the hybrid convolution output to overcome the accuracy degradation problem caused by long convolutional layers and poor generalization performance of a single network structure. An additional branch performs simple linear iterative clustering and entropy rate superpixel (ERS) segmentation, which are sequentially implemented on the HSI to solve the under-segmentation problem. This strategy is important, as dynamically calculating the tuning parameters for feature segmentation maps increases the number of the multiscale GCN layers and fully extracts contextual spatial information. Experiments on three public datasets, Indian Pines, Kennedy Space Center, and the University of Pavia, demonstrate that the proposed framework achieves the optimal OA, AA, and Kappa coefficients. The source code is available at https://github.com/henulx/HDECGCN-Framework.**

***Index Terms*— Double superpixel segmentation, dynamic tuning parameters, hybrid 3-D–2-D convolutional neural networks (CNNs), hyperspectral image (HSI) classification, multiscale graph convolutional networks (GCNs), residual nonlocal block.**

I. Introduction

HYPERSPECTRAL remote sensors are commonly referred to as imaging spectrometers. The imaging spectroscopy technology used by the spectrometer obtains target spatial information and provides dozens or even hundreds of narrow-band spectral information for each pixel. The data obtained by the imaging spectrometer include 2-D spatial and 1-D spectral information, and the generated 3-D data cube is called a hyperspectral image (HSI) [1]. Due to the spatial-spectral joint information in HSIs, processing HSIs and extracting features of interest from the images at a pixel level is possible. Owing to these advantages, HSIs have been widely used in environmental monitoring [2], precision agriculture [3], geological exploration [4], target detection [5], and other fields closely related to human production and life.

Classifying ground objects in HSIs relies on the different spectral information of the various ground objects, which is used to classify different ground object categories on a pixel level [6]. Therefore, HSIs achieve higher classification accuracy (CA). The classification algorithms for HSIs are divided into nondeep and deep learning algorithms, both aimed at obtaining discriminative information of interests in HSIs [7]. Nondeep learning HSIs with good classification performance often use spatial-spectral joint features. For example, Jiang et al. [8] proposed an unsupervised multiscale superpixel PCA HSI classification algorithm based on texture similarity (MSuperPCA). Although the spatial-spectral joint nondeep learning HSI classification algorithm has stronger representation and generalization capabilities, it can extract deeper image features and achieve good classification results. However, it mostly relies on manual feature extraction, involving a large amount of prior knowledge and parameter settings from experts. Processing becomes more difficult when faced with high spatial size and large sample-size datasets [9].

In recent years, deep learning neural networks have been widely applied in remote sensing (RS) data processing [10], [11], [12], [13]. Yu and Jia [14] introduced an effective 1D-CNN architecture to improve its discriminative ability for HSI classification, and Makantasis et al. [15] proposed a deep supervised learning algorithm based on 2D-CNN to classify HSIs. Nevertheless, 1D-CNN only extracted spectral information and 2D-CNN lacked the extraction of spatial-spectral joint information. Zhong et al. [16] proposed a spectral-spatial residual network for HSI classification (SSRN). The residual blocks in SSRN used identity mapping to connect each other to the 3-D convolutional layer, compared to SSRN.

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Roy et al. [17] used a hybrid spectral convolutional neural network (CNN) to reduce the complexity of the whole model (HybridSN). However, due to their long network structure, those two direct CNN architectures cannot effectively utilize the relationships between HSI features in the spatial-spectral domain [16], [17]. Compared with the HybridSN [17], which uses a common pooling layer, Zheng et al. [18] proposed mixed CNNs and covariance pooling to extract more important second-order information (MCNN-CP). Additionally, Lei et al. [19] embedded an additional convolutional layer before the capsule layer and introduced a nonlocal block between the convolutional layers, capturing the long-distance dependencies of HSIs. Chen et al. [20] rethought the feature extraction of HSI from an anisotropic perspective and proposed a novel model called a grid network. Recently, attention networks combining spatial-spectral methods have been used for HSI classification. For instance, Li et al. [21] proposed a dual branch dual attention mechanism CNNs for HSI classification (DBDA). The dual branch network still employs ordinary 3D-CNN to extract features that contain significant spatial information redundancy. Different from CNNs, transformers are one of the frontier network architectures due to the use of self-attention techniques, which have already proved their advantages in processing sequential data [22]. He et al. [23] proposed a transferring spatial-spectral transformer to capture sequence spectral relations and used label smoothing to complete HSI classification. Hong et al. [24] rethought HSIs from the sequence perspective and first proposed a transformer-based backbone network called SpectralFormer for HSI classification. The SpectralFormer devised two simple but effective techniques, i.e., groupwise spectral embedding and cross-layer adaptive fusion, to learn spectrally local sequence information from neighboring bands and convey memory-like components from shallow to deep layers, respectively. However, with the large-scale growth of RS data from different platforms, RS AI models urgently require a large number of parameters, and researchers have designed various basic models. Hong et al. [25] first proposed SpectralGPT, a universal spectral RS basic model, trained on over one million spectral images and over 600 million parameters, it achieved intelligent processing of spectral RS images.

Recently, graph convolutional networks (GCNs) have been widely used in node classification recommendation systems [26]. Besides, GCNs can also model the spatial context structure of HSIs. For example, Qin et al. [27] first proposed a new semi-supervised learning framework based on spatial-spectral joint GCNs, connecting the internal information of labeled and unlabeled samples to alleviate the problem of sample shortage. Chen et al. [28] presented a novel GCN-based method, called local aggregation and global attention block by using a spectral-induced aligned superpixel segmentation technique and achieving high-accuracy classification results. Wan et al. [29] designed a multiscale dynamic GCN (MDGCN) using dynamic convolution operation. Similarity measurement between pixels can be updated by fusing the current feature embedding. However, when calculating the adjacency matrix, constant tuning parameters are used without considering the weight of adjacent hyperpixels, which inevitably misclassifies some pixels, leading to the inability to optimize the graph topology. MDGCN used simple linear iterative cluster (SLIC) segmentation, inevitably leading to under-segmentation. Liu et al. [30] proposed an enhanced graph encoder and decoder to propagate features between image pixels and graph nodes in order to alleviate the incompatibility of data representation structures between CNNs and non-Euclidean GCNs for European data (CEGCN). However, when facing sparse and small-scale training sets, the CEGCN lacks complementary information between pixels and superpixels. Dong et al. [31] proposed a weighted feature fusion of CNN and graph attention network (WFCG), which proved to be complementary. Zhou et al. [32] proposed a novel multiple-feature fusion model termed attention multihop graph and multiscale CNN (AMGCFN). The complementary problem also can be improved from an RS data-driven perspective. For instance, Wu et al. [12] and Hong et al. [33] studied the use of dual-branch CNN architectures to combine features from two different modalities (e.g., hyperspectral, LiDAR, and SAR), the two multimodal classification network makes the features of different modalities obtained by the network extractor fused in a more compact way. Further, Hong et al. [34] proposed a multimodal cross-city RS dataset and a high-resolution domain adaptation network for semantic segmentation task. In summary, the combined use of different modality data can extract more accurate and comprehensive pixel-level information.

From the algorithm-driven perspective, this article proposes a heterogeneous dual-enhanced network utilizing CNNs-GCNs (HDECGCN). Fig. 1 illustrates HDECGCN, which is comprised of two strengthening sub-networks. Specifically, the CNN subnetwork is a branch that performs nonpositioning operations on the image's spectral bands to overcome intraclass similarity problems. Then, a three-channel stack extracts deep and shallow spectral information at the end of a $64 \times 1 \times 1$ convolution kernel, we perform group convolution on both sides of the three-channel stack to overcome overfitting caused by too many parameters. Meanwhile, the Mish activation function [35] is used for all 2-D convolutions to enhance the network's generalization performance. Additionally, the residual nonlocal block at the output end of the 2-D convolution compensates for the CNN's inability to capture long-distance dependency. An enhanced GCN is the second subnetwork, which reperforms entropy rate superpixel (ERS) segmentation on the segmented image obtained by SLIC to overcome the under or over-segmentation problem. Note that using dynamic tuning parameters [36] increases the number of layers in the GCNs to obtain contextual information at different levels. Finally, the decision fusion algorithm optimizes the CA obtained by two subnetworks to maximize CA, thereby achieving fully complementary spatial-spectral joint features at the pixel and superpixel levels. We summarize the contributions as follows.

1) A hybrid three-channel stack CNNs with residual nonlocal block based on the group convolution is proposed (Section III-A). We introduce group convolution on

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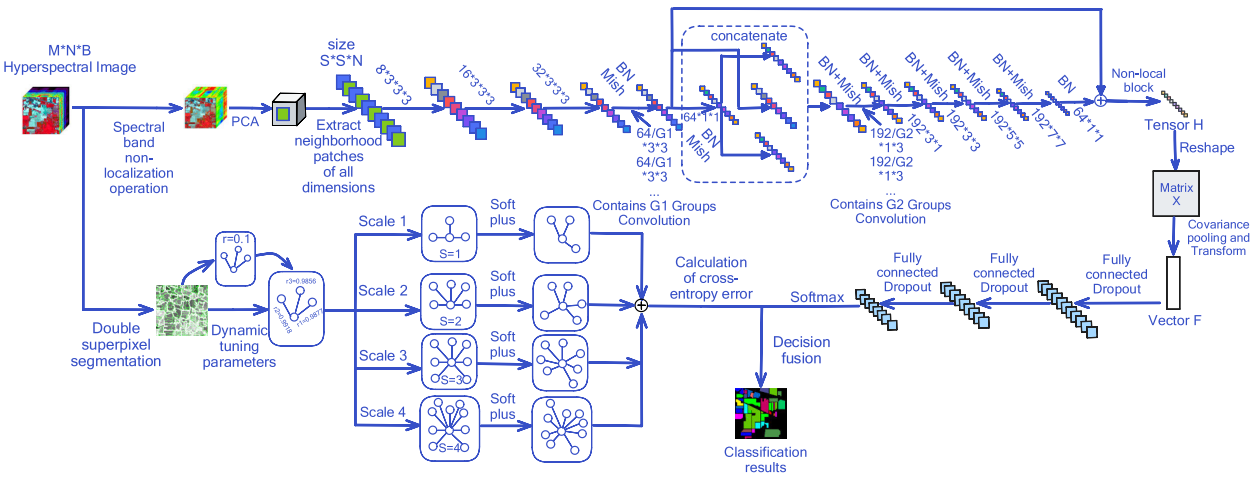


Fig. 1. HDECGCN framework. The enhanced CNN subnetwork obtains the vector F through covariance pooling and transforms, followed by three fully connected layers to export the classification map. After calculating the tuning parameters of the initial adjacency matrix of the enhanced GCNs subnetwork, we construct a graph structure of four scales and use the SoftPlus function for the generated graph of each scale. After that, each generated graph is accumulated, and the cross-entropy error is adopted to penalize the difference between the network output and the labels of the original labeled examples to obtain a ground feature classification map. Finally, the decision fusion algorithm obtains the final classification result graph of the HDECGCN framework.

the left and right sides of the three-channel stack to overcome overfitting caused by too many parameters. Besides, A residual nonlocal block is added to the output end of the 2-D convolution of the enhanced hybrid CNNs to capture the HSI's long-distance dependency and solve the CA degradation caused by long convolution layers [16], [17], [18].

2) To optimize the nodes of GCNs. Using ERS segmentation to achieve secondary segmentation of the feature images obtained from SLIC segmentation (Section III-B). Specifically, the entire HSI is segmented into uniform regions with similar reflectance features to solve the under-segmentation problem [29], [30], [31], [32].
3) To optimize the edges of GCNs. We use the average standard deviation to calculate the tuning parameters and reflect the weights between neighboring superpixels (Section III-B). The improved tuning parameters are employed to calculate the adjacency matrix between adjacent superpixels, and the new adjacency matrix is used to calculate the multiscale dynamic GCNs by increasing the number of scales in the network and mining rich spatial contextual information. This improvement compensates for MDGCN neglecting the neighborhood superpixel weights [29].
4) To solve the problem of the high running time of hybrid frameworks in the classification of HSIs (PaviaU and KSC) [30], [31], [32], we use a decision fusion algorithm with equal voting strength to obtain the final classification result graph, achieving subnetwork compatibility with linear time complexity (Section III-C).

II. Related Works

This section reviews the mature technologies related to the innovation points of this article.

A. Nonlocal Block

Capturing long-range dependencies is of central importance in deep neural networks. For image data, convolutional operations process a local neighborhood, either in space or time; thus, long-range dependencies can only be captured when these operations are repeatedly applied by propagating signals progressively through the data [37]. However, repeating local operations has several limitations: it is computationally inefficient and causes optimization difficulties that must be carefully addressed [38]. Wang et al. [39] proposed nonlocal operations for capturing remote dependencies in CNN. Besides, a nonlocal block has been proposed based on a nonlocal mean algorithm, which calculates the weighted mean of all pixels in an image [40]. Later, block matching 3D (BM3D) was also based on nonlocal mean filtering [41].

The general nonlocal operations in neural networks are defined as [40]

$$\boldsymbol{y}_i = \frac{1}{C(\boldsymbol{x})} \sum_{\forall_j} f\left(\boldsymbol{x}_i, \boldsymbol{x}_j\right) g\left(\boldsymbol{x}_j\right) \tag{1}$$

where i is the index of an output position (in space, time, or spacetime) whose response is to be computed, and j is the index that enumerates all possible positions. $\boldsymbol{x}$ is the input signal (image, sequence, video; often their features), and $\boldsymbol{y}$ is the output signal of the same size as $\boldsymbol{x}$. A pairwise function f computes a scalar between i and all j. The unary function g computes a representation of the input signal at the position j, and the response is normalized by a factor $C(\boldsymbol{x})$. The results of nonlocal operations are not sensitive to the selection of functions f and g [39].

We encapsulate the nonlocal operations in formula (2) and merge them into the existing framework. Hence, a nonlocal block is defined as

$$\boldsymbol{z}_i = \boldsymbol{W}_z \boldsymbol{y}_i + \boldsymbol{x}_i \tag{2}$$

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where "$+\boldsymbol{x}_i$" denotes a residual connection and $\boldsymbol{W}_z$ is a weight matrix to be learned. In HSI classification tasks, the nonlocal behavior of the nonlocal block can simulate the long-distance dependence of HSIs [19].

B. ERS Segmentation

The basic idea of the ERS segmentation proposed by Liu et al. [42] is to treat superpixel segmentation as a clustering problem and obtain the segmentation results by solving the maximum value of a certain objective function. ERS introduces a new objective function that includes the entropy rate term of random walks on the graph and the equilibrium term of the clustering distribution. The former can ensure compact and uniform clustering, while the latter ensures that each superpixel region segmented is the same size.

Specifically, ERS first constructs a very large undirected graph $G = (V, E)$, where the set of vertices V represents the image pixels and the edge set E represents the relationships between pixels. The algorithm's goal is to select a subset α of the edge set, $\alpha \subseteq E$. The resulting graph is $G' = (V, \alpha)$, containing K connected subsets. Liu et al. [42] constructed a clustering objective function based on a greedy algorithm to solve the resulting graph, which has been widely applied in HSI classification. Wang et al. [43] proposed an HSI classification method based on spectral texture kernels, and Liu et al. [44] developed an HSI classification algorithm based on PCA domain superpixel singular spectrum analysis.

C. Weighted Label Propagation

Wang and Tsotsos [45] proposed a dynamic label propagation (DLP) algorithm. Two parameters must be set when classifying HSI to regularize the label correlation and identity matrix. However, since the two parameters are artificially specified, these cannot easily adapt to HSIs with different spatial coverage and resolution. Thus, the generalization performance of the DLP algorithm gradually declines. Therefore, Jia et al. [36] proposed a weighted label propagation algorithm based on superpixel level for HSI classification, using standard deviation to reflect the degree of dispersion within superpixels. The standard deviation of superpixels σ_k can be defined as

$$\boldsymbol{M}_k^b = \frac{1}{m_k}\sum_{i,j}\Re_{(i,j,b)}, \quad \boldsymbol{S}_{(i,j)} = k \;\; k = 1, \ldots, K, \;\; b = 1, \ldots, N \tag{3}$$

$$\sigma_k = \frac{1}{N}\sum_{b=1}^{N}\sqrt{\frac{1}{m_k}\sum_{i,j}\left(\Re_{(i,j,b)} - \boldsymbol{M}_k^b\right)^2}\boldsymbol{S}_{(i,j)} = k \tag{4}$$

where N is the number of spectral bands, m_k is the number of pixels inside the kth superpixel $\boldsymbol{S}_{(i,j)}$, $\Re_{(i,j,b)}$ represents any pixel where $\boldsymbol{S}_{(i,j)}$ in the bth band, the bth element $\boldsymbol{M}_k^b, b = 1, \ldots, N$, of the feature mean vector $\boldsymbol{M}_k \in R^B$ corresponding to the kth superpixel $\boldsymbol{S}_{(i,j)}, k = 1, \ldots, K$, is computed by formula (3). So, the average standard deviation of the entire superpixel image $\boldsymbol{S}$ is $\boldsymbol{\sigma} \in R^K$

$$\boldsymbol{\sigma} = [\sigma_1, \sigma_2, \ldots, \sigma_K]^T. \tag{5}$$

A smaller standard deviation σ_K represents a higher uniformity of the superpixel structure. On the contrary, the larger the standard deviation, the stronger the heterogeneity within the superpixel. In label propagation algorithms, superpixels with low standard deviation should be given greater weight. Therefore, a weight function was designed to obtain the weight parameters $\boldsymbol{w} \in R^K$

$$\boldsymbol{w} = \exp(-\boldsymbol{\sigma}). \tag{6}$$

The weight parameter $\boldsymbol{w}$ decreases as the standard deviation of weights increases. The weight parameters are automatically calculated for the weighted label propagation algorithm [32], enhancing the generalization ability for different HSIs without requiring manual settings to extract all superpixels-based information.

III. Proposed Method

A. A Hybrid Residual Channel Stack Network Based on Attention Covariance Pooling

Assume an original HSI and denote it as $\boldsymbol{H} \in R^{X*Y*N}$, where X and Y are two spatial dimensions, and N is the number of bands. The spectral features of HSIs generally have interclass similarity issues, which is not conducive to using spectral features for classification. For HSI classification based on CNNs, it is challenging for the convolution kernel with a limited size to learn the dependence between the distant bands among the bands. Thus, Gao et al. [46] proposed a spectral band nonlocalization (SBNL) operation to connect the ith band B_i with the $N-i$th band B_{N-i}, overcoming interclass similarity.

This article uses SBNL on the HSI by inserting nonadjacent long-distance bands between adjacent bands and performing PCA on the reconstructed bands to obtain a feature image $\boldsymbol{D}_{\mathrm{NL}}$. $\boldsymbol{D}_{\mathrm{NL}}$ is divided into small overlapping 3-D blocks, whose truth labels are determined by the labels of the central pixel. Then, 3-D neighboring patches $\boldsymbol{P} \in R^{S*S*N}$ are created from $\boldsymbol{D}_{\mathrm{NL}}$, centered at the spatial location (β, ξ), covering the $S \times S$ window and all bands, using the patch $\boldsymbol{P}$ as input data.

For the network structure depicted in Fig. 1, the dimensions of the 3-D convolutional kernels are $8 \times 3 \times 3 \times 3$, $16 \times 3 \times 3 \times 3$, and $32 \times 3 \times 3 \times 3$, respectively, serving as the size of the first, second, and third convolutional layer. Note that filter $16 \times 3 \times 3 \times 3$ represents all 16 3-D kernels with dimensions of $3 \times 3 \times 3$. Using three 3-D convolutions preserves the spectral information of the image at the end. After performing the 3-D convolution, Fig. 2 shows a schematic of the three-channel stack, which contains two 2-D convolution kernels with BN and Mish activation functions. The filter dimensions are $64 \times 1 \times 1$ and $\{(64/G_1) \times 3 \times 3, (64/G_1) \times 3 \times 3, \ldots,\}$, serving as the size of each 2-D convolution layer, $G_1 \in N^*$ is the number of groups for group convolution. Let the feature of the output of the convolution $64 \times 3 \times 3$ be $\boldsymbol{F}_1$, and the feature of the output of the convolution $64 \times 1 \times 1$ be $\boldsymbol{F}_2$. We perform a dual connection operation on $\boldsymbol{F}_1$ by placing $\boldsymbol{F}_2$ on its left and right sides, with $\boldsymbol{F}_1$ acting as the axis.

For the convolutional kernels, $64 \times 3 \times 3$, 64 represents the number of 2-D convolutional kernels, and 3×3 is the

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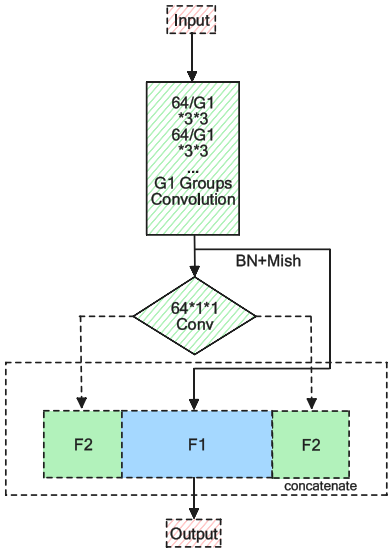


Fig. 2. Structure diagram of the three-channel stack.

spatial dimension. The three-channel stack can capture spectral features with discriminative performance, the 3 × 3 sized convolution kernels are used for deep spectral extraction from HSIs, and the two 1 × 1 sized convolution kernels extract shallow features. This architecture achieves the stacking of deep-shallow spectral information.

Next, we introduce a residual module at the output of the 64 × 3 × 3 convolutional layer. The motivation is that the CA decreases with the increase of the convolutional layers. This phenomenon stems from the fact that the representation capacity of the CNNs is too high compared with the relatively small number of training samples with the same regularization settings [16]. So, this problem can be alleviated by adding shortcut connections between every other layer to build a residual block [47]. This architecture enables higher-level gradients to rapidly propagate back to the lower layers, facilitating and regularizing the model training process [16]. However, residual structures cannot effectively express information about the target of interest and suppress irrelevant information [19]. As depicted in Fig. 3, we propose a covariance pooling structure based on an attention mechanism, which introduces a nonlocal block at the output of the residual model to capture the dependencies between long-distance pixels of HSIs. Compared to ordinary pooling, it can extract more important second-order information between long-distance pixels and prevent overfitting.

According to Fig. 3, we propose a residual nonlocal block structure based on the three-channel stack. In order to distinguish the spatial information in different spectral bands and maintain high CA, we construct a six-layered 2D-CNN based on the Mish and place it after the three-channel stack, which contains convolution kernels with dimensions $\{(192/G_2) \times 1 \times 3, (192/G_2) \times 1 \times 3, \ldots,\}$, $192 \times 3 \times 1$, $192 \times 3 \times 3$, $192 \times 5 \times 5$, $192 \times 7 \times 7$, and $64 \times 1 \times 1$, $G_2 \in N^*$ is the number of groups for group convolution. Table I reports the type, output shape, and number of parameters of each CNN layer of the hybrid residual channel stack network based on attention covariance pooling (HR-ACPCNN) on the PaviaU with a window size of 11 × 11.

TABLE I

TYPE, OUTPUT SHAPE, AND PARAMETERS OF EACH LAYER OF THE HR-ACPCNN ON THE PAVIAU (WINDOW SIZE OF 11 × 11)

Layer(type)	Output Shape	# Parameter
input_1(InputLayer)	(11,11,30,1)	0
conv3d_1(Conv3D)	(11,11,30,8)	224
conv3d_2(Conv3D)	(11,11,30,16)	3472
conv3d_3(Conv3D)	(11,11,30,32)	13856
reshape_1(Reshape)	(11,11,960)	0
conv2d_1(Conv2D)	(121,64)	276544
conv2d_2(Conv2D)	(11,11,64)	4160
conv2d_3(Concatenate)	(11,11,64)	284864
conv2d_4(Conv2D)	(11,11,192)	37056
conv2d_5(Conv2D)	(11,11,192)	110784
conv2d_6(Conv2D)	(11,11,192)	331968
conv2d_7(Conv2D)	(11,11,192)	921792
conv2d_8(Conv2D)	(11,11,192)	1806528
conv2d_9(Conv2D)	(11,11,64)	12352
Total Trainable Parameters in All Mixed Convolutional Layers: 3803600		

B. Graph Convolutional Network for Double Hyperpixel Segmentation Based on Multiscale Dynamic Tuning Parameters

Wan et al. [29] and Liu et al. [30] used SLIC for preprocessing an original HSI $\boldsymbol{H} \in R^{X*Y*N}$ used as an input image. They divided $\boldsymbol{H}$ into several uniformly sized superpixels. However, SLIC can cause under-segmentation, as the ability to distinguish class differences decreases when pixels are very close. Since ERS segmentation can divide images into compact, uniform, and interclass differences balanced clusters [42], we still use SLIC to segment the entire image into several superpixels and then employ ERS [42] for secondary segmentation of the feature image. Fig. 4 illustrates a schematic of the dual segmentation.

In HSI classification methods based on GCNs, the graph structure is usually constructed using pixels or superpixels as nodes. One drawback of traditional GCN is that the graph is fixed throughout the convolution process. If the input graph structure is inaccurate, CA will be reduced. Thus, MDGCN can gradually be refined during the convolution process to address this issue. The main idea is to find an improved graph by fusing the information of current data embeddings and the graph in the previous layer [29]. According to the nuclear fusion in [45] and [48], the initial adjacency matrix should carry the inherent structure among the input data. Based on the two technologies, the adjacency matrix $\boldsymbol{A}^{(l+1)}$ can be dynamically updated as

$$\boldsymbol{A}^{(l+1)} \leftarrow \boldsymbol{A}\left(\boldsymbol{A}^{(l)} + \theta \boldsymbol{H}^{(l)} \boldsymbol{H}^{(l)^T}\right)\boldsymbol{A}^T + \beta^{(l)} \boldsymbol{I} \tag{7}$$

where $\boldsymbol{A}^{(l)}$ represents the adjacency matrix, $\boldsymbol{H}^{(l)}$ is the output matrix, $\boldsymbol{A}$ is the initial adjacency matrix, $\boldsymbol{I}_m$ stands for the identity matrix, $\beta^{(l)}$ controls the relative importance of $\boldsymbol{I}_m$, L represents the index of the layer in which it is located, and θ is the weight assigned to the kernel $\boldsymbol{H}^{(l)} \boldsymbol{H}^{(l)^T}$.

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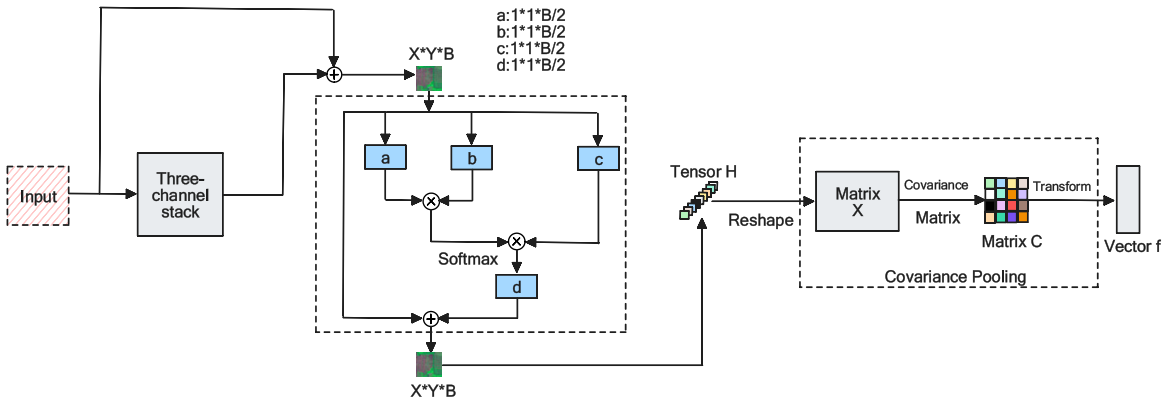


Fig. 3. Covariance pooling structure based on the mixed residual attention mechanism.

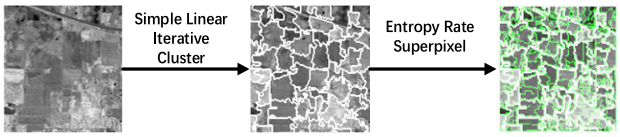


Fig. 4. Schematic of dual superpixel segmentation (SLIC-ERS).

In order to calculate the adjacency matrix $\boldsymbol{A}^{(l+1)}$ in formula (7), the initial adjacency matrix $\boldsymbol{A}^{(1)}$ is calculated as follows:

$$\boldsymbol{A}_{i,j}^{(1)} = e^{-\gamma \|\boldsymbol{x}_m - \boldsymbol{x}_n\|^2} \tag{8}$$

where γ is the tuning parameter, $\boldsymbol{x}_m$ represents the mth super-pixel, and $\boldsymbol{x}_n$ is the neighboring superpixel of $\boldsymbol{x}_m$. In MDGCN [29], $\gamma = 0.1$ and does not change with dataset changes. If $\boldsymbol{x}_n$ and $\boldsymbol{x}_m$ are distributed between two adjacent features, and γ is set to a fixed value, the inconsistent weights of neighboring superpixels are not considered, resulting in a decrease in the smoothing accuracy of edge information.

To solve this problem, inspired by Jia et al. [36], we construct graphs at different scales based on dual superpixel segmentation and apply the concept of a weighted label propagation algorithm to GCNs. That is, we use the standard deviation to construct dynamic tuning parameters $\boldsymbol{\gamma}_D$ in the adjacency matrix to reflect the different weights between neighboring superpixels. The calculation method for the standard deviation σ_k of superpixels is consistent with formulas (3) and (4) in Section II-C. From this, the average standard deviation of the dual superpixel segmentation image $\boldsymbol{S}_D$ is obtained $\boldsymbol{\sigma} \in R^K$

$$\boldsymbol{\sigma} = \left[\sigma_1, \sigma_2, \ldots, \sigma_{K_p}\right]^T \tag{9}$$

where K_p represents the number of pixels generated by dual superpixel segmentation. Similarly, a smaller standard deviation indicates superpixel structure uniformity, which meets the expectations of the dual superpixel segmentation algorithm. A larger standard deviation results in larger differences within the superpixels. Therefore, we design a weighted function to obtain the dynamic tuning parameter $\boldsymbol{\gamma}_D$ in formula (8), defined as

$$\boldsymbol{\gamma}_D = \exp(-\boldsymbol{\sigma}). \tag{10}$$

The dynamic tuning parameter $\boldsymbol{\gamma}_D$ decreases with the increase of $\boldsymbol{\sigma}$, and the weight of superpixels decreases with the increase of standard deviation, corresponding to the uniformity of the dual superpixel segmentation graph structure. It can also automatically calculate dynamic tuning parameters and set different parameters for different datasets to improve generalization. In order to improve the quality of the convolution graph, the adjacency matrix between nonadjacent hyperpixels is set to 0

$$\boldsymbol{A}_{ij}^{(l)} = \begin{cases} \boldsymbol{A}_{ij}^{(l)}, & \text{if } \boldsymbol{x}_i \in Nei(\boldsymbol{x}_j) \text{ or } \boldsymbol{x}_j \in Nei(\boldsymbol{x}_i) \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

C. Heterogeneous Dual Enhanced CNNs-GCNs Network

The HDECGCN framework comprises two sub-networks: HR-ACPCNN and graph convolutional network for double hyperpixel segmentation based on multiscale dynamic tuning parameters (MDT-DSGCN), each with advantages. HR-ACPCNN absorbs the HybridSN [17] and combines the complementary information of spatial-spectral and spectral in the form of 3-D–2-D convolutions. Besides, the Mish and residual nonlocal blocks are introduced to solve the problem of reduced CA due to the long-mixed convolution layers. Therefore, the HR-ACPCNN can extract rich spatial-spectral joint information compared to SSRN [16] and MCNN-CP [18]. It was observed that CNN-based networks learn pixel-level features in small regular regions, and the generated land-cover classification map is pixel-level oriented.

The MDT-DSGCN captures multiscale spatial-spectral joint information by constructing scale maps of different neighborhoods and calculating dynamic tuning parameters based on the dual superpixel segmentation. Compared to MDGCN [29], it increases the number of scale maps and captures more spatial-spectral joint information. Additionally, the GCN branch-based joint superpixel segmentation can model large irregular regions to generate superpixel-level features, and the corresponding generated land-cover classification map is oriented toward the superpixel level. Inspired by the concept of combining CNNs-GCNs branches using the minibatch GCN [49] and CEGCN [30], and unlike fusing them [30], [49], the minibatch GCN [49] must select the optimal fusion method manually. Therefore, we use decision fusion strategies

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to improve the CA of the land-cover classification maps output by the two branches [8]. There are two HSI classification maps of ground objects, and next, a decision fusion strategy is used to aggregate the results. This article leverages the majority voting-based decision fusion strategy due to its insensitivity to inaccurate estimates of posterior probabilities

$$l = \arg\max_{i\in\{1,2,\ldots,G\}} N(i), \quad N(i) = \tfrac{1}{2}I(l_1 = i) + \tfrac{1}{2}I(l_2 = i) \quad (12)$$

where l is the class label from one of the g possible classes for the test pixel, $N(i)$ represents the number of times that class i is predicted in the bank of two classifiers, and I denotes the indicator function. In formula (12), we set the equal voting strength to 0.5.

D. Motivation for the Proposed Method

We aim at developing a novel and generic CNNs-GCNs-based framework (i.e., HDECGCN) with a focus on generating complementary spatial-spectral joint information at the superpixel-pixel levels, making it well-applicable to the fine accuracy and less running time. However, CNNs cannot directly model long-range spatial relationships in HSIs [29], [49]. And the graph structure of traditional GCNs is not optimal, specifically, the SLIC may result in under-segmentation in regions with abundant terrain features [30], [31], [32], and the relationship between neighboring superpixel blocks has been determined manually and fixedly [29]. To address the two issues, we use the ERS for secondary segmentation to make the superpixels uniform and compact, and the average standard deviation is employed to calculate the tuning parameters and reflect the weights between neighboring superpixels. Although the MDT-DSGCN is complete, the lack of spatial-spectral joint information in short-range regions and the errors inherent in superpixel segmentation are still an objective reality, motivated by the CNNs for pixel-level. We mix CNNs and GCNs, however, CNNs which in series suffer from overfitting due to excessive parameters and lack of capturing contextual information of HSIs [16], [17], [18]. A hybrid three-channel stack CNNs with residual nonlocal block based on the group convolution is proposed to solve the issues (HR-ACPCNN). In order to make the two subnets compatible and reduce the running time caused by the joint encoder-decoder [30], [31], [32], inspired by MSuperPCA [8], we use a decision fusion algorithm with equal voting strength to obtain the final classification result graph, achieving subnetwork compatibility with linear time complexity. Finally, HDECGCN was born.

IV. Experiments and Results

A. Data Description

To demonstrate the effectiveness and generalization ability of the HDECGCN framework, three commonly used public datasets from different imaging platforms are adopted: Indian Pines, Kennedy Space Center (KSC), and PaviaU. Table II presents their basic information, where the related sensors, band information, label ratios, spatial size, number of samples, classes, and data acquisition years are presented. The pseudo-color images and the corresponding ground-truth maps of the three datasets are also presented in Figs. 5–7.

TABLE II
Detailed Information of Three Hyperspectral Datasets

	Indian Pines	PaviaU	KSC
Sensor	AVIRIS	ROSIS	AVIRIS
Bands	200	103	176
Label Ratio	48.75%	20.62%	1.7%
Spatial Size	145×145	610×340	512×614
Num. of Samples	10249	42776	5211
Num. of Classes	16	9	13
Year	1992	2001	1996

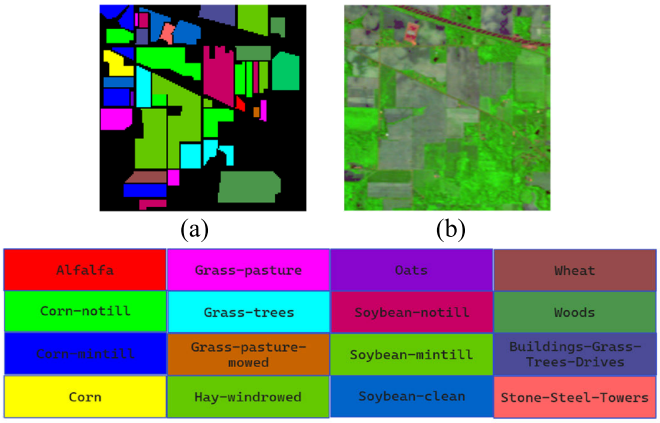


Fig. 5. Indian Pines dataset (a) ground-true map, (b) false-color image, and (c) detailed types of land cover and color-coded.

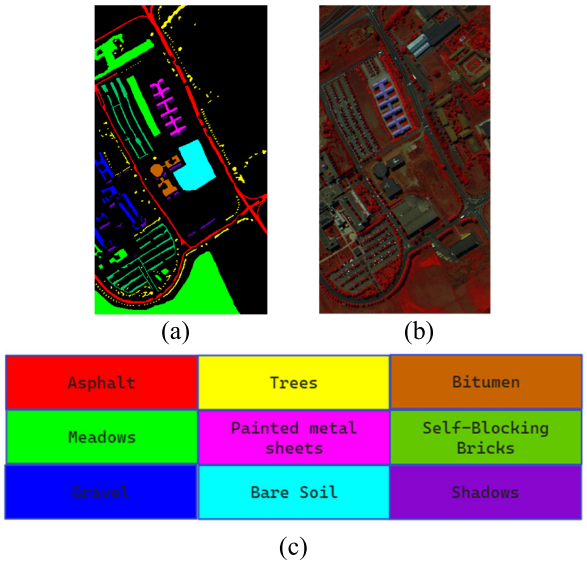


Fig. 6. PaviaU dataset (a) ground-true map, (b) false-color image, and (c) detailed types of land cover and color-coded.

B. Experimental Setup

All trials are conducted on an Intel i7-12700KF 3.60 GHz processor with 64 GB RAM and an NVIDIA GeForce RTX 3070 Ti with 8 GB RAM. All methods are implemented on Python 3.8 using the Tensorflow 2.4 library and MATLAB R2021a. Tables III–V report the number of labeled and un-labeled pixels of various classes for the three datasets. We chose samples of moderate size, and limited samples

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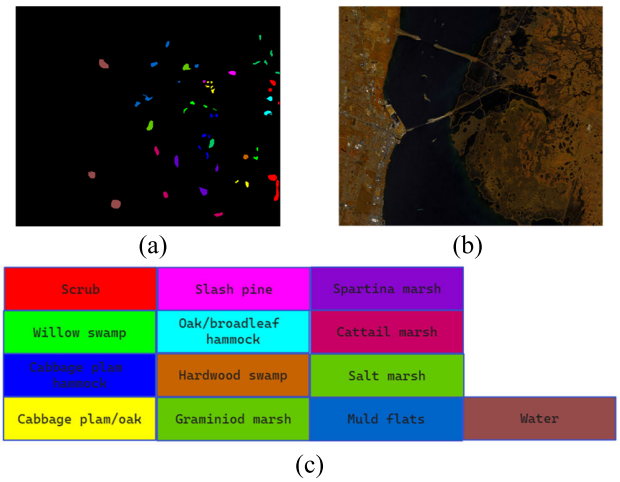


Fig. 7. KSC dataset (a) ground-true map, (b) false-color image, and (c) detailed types of land cover and color-coded.

TABLE III
LAND COVER CATEGORIES AND THE DATASET DIVISION FOR EACH CATEGORY OF INDIAN PINES DATASET

Class	Name	Training	Testing
1	Alfalfa	5	41
2	Corn-notill	143	1285
3	Corn-mintill	83	747
4	Corn	24	213
5	Grass-pasture	48	435
6	Grass-trees	73	657
7	Grass-pasture-mowed	3	25
8	Hay-windrowed	48	430
9	Oats	2	18
10	Soybean-notill	97	875
11	Soybean-mintill	245	2210
12	Soybean-clean	59	534
13	Wheat	20	185
14	Woods	126	1139
15	Bldg-Grass-Trees-Drives	39	347
16	Stone-Steel-Towers	9	84
	Total(Percentage)	1024(10%)	9225(90%)

TABLE IV
LAND COVER CATEGORIES AND THE DATASET DIVISION FOR EACH CATEGORY OF PAVIAU DATASET

Class	Name	Training	Testing
1	Asphalt	398	6233
2	Meadows	1119	17530
3	Gravel	126	1973
4	Trees	184	2880
5	Painted metal sheets	80	1265
6	Bare Soil	301	4728
7	Bitumen	80	1250
8	Self-Blocking Bricks	221	3461
9	Shadows	57	890
	Total(Percentage)	2566(6%)	40210(94%)

are used as the training set for each dataset in subsequent experiments to represent the performance of the HDECGCN.

In the HDECGCN framework, the two branch networks are implemented via Tensorflow using the Adam optimizer,

TABLE V
LAND COVER CATEGORIES AND THE DATASET DIVISION FOR EACH CATEGORY OF KSC DATASET

Class	Name	Training	Testing
1	Scrub	46	715
2	Willow swamp	15	228
3	Cabbage plam hammock	15	241
4	Cabbage palm/oak	15	237
5	Slash pine	10	151
6	Oak/broadleaf hammock	14	215
7	Hardwood swamp	6	99
8	Graminoid marsh	26	405
9	Spartina marsh	31	489
10	Cattail marsh	24	380
11	Salt marsh	25	394
12	Muld flats	30	473
13	Water	55	872
	Total(Percentage)	312(6%)	4899(94%)

TABLE VI
NUMBER OF EPOCHS, LEARNING RATE, SPATIAL SIZE, BATCH SIZE, AND GROUP CONVOLUTIONS IN THE HDECGCN ON THE THREE DATASETS

	Indian Pines	KSC	PaviaU
Epochs in HR-ACPCNN	50	50	50
Epochs in MDT-DSGCN	4000	4000	7000
Learning Rate	0.001	0.001	0.001
Spatial Size	11×11	11×11	11×11
Batch Size	256	256	256
Num. of group convolutions	1	1	64

where the learning rate is 0.001 for all three datasets. The MDT-DSGCN maintains the same parameters for all experimental datasets because the GCN-based HSI classification algorithm does not require a deep structure to achieve high CA [29]. Specifically, four neighborhood scales are employed for graph construction to incorporate multiscale spectral-spatial information into MDT-DSGCN. After trial-and-error experiments at each scale, we set up two GCNs and 20 hidden units. Besides, the number of training epochs is 4000 for Indian Pines and KSC, while 7000 for the PaviaU. In the HR-ACPCNN, we use the same spatial size of 11 × 11 pixels as inputs. The number of training epochs and the batch size are 50 and 256, respectively, for the three datasets. We use 256, 128, and 16 neural units for the three fully connected layers, and the dropout of the layers is 0.4. Table VI summarizes the three datasets' epoch values, learning rates, and spatial and batch sizes. The average accuracy (AA), overall accuracy (OA), Kappa coefficient, and running time are used to evaluate the performance of HDECGCN.

In order to evaluate the classification performance of the HDECGCN, eight state-of-the-art methods have been selected for comparison, including MSuperPCA [8], SSRN [16], HybridSN [17], MCNN-CP [18], DBDA [21], MDGCN [29], CEGCN [30], WFCG [31], and AMGCFN [32]. For a fair comparison, all parameters of the competitor models adopt the optimal parameter values given in the literature.

C. Parameter Analysis

The optimal parameter values for HDECGCN are determined by performing HSI classification and modifying each

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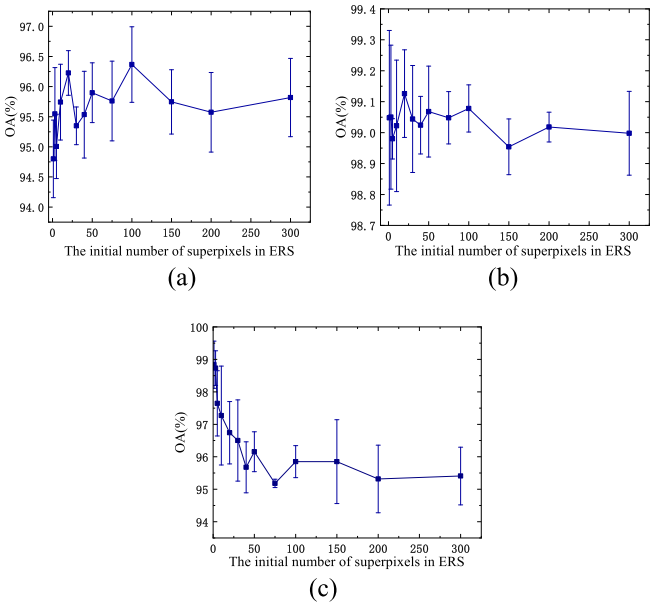


Fig. 8. Influences of the number of superpixels S_I on the OA (%) of the three datasets (a) Indian Pines, (b) PaviaU, and (c) KSC.

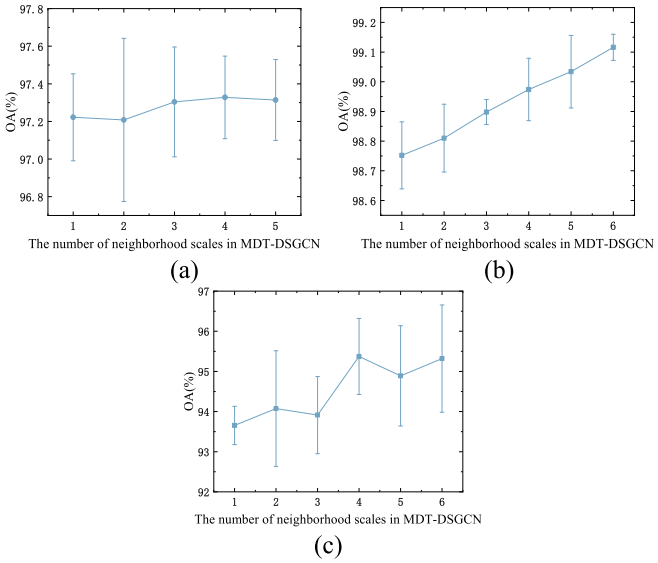


Fig. 9. Influences of the number of neighborhood scales C on the OA (%) of the three datasets (a) Indian Pines, (b) PaviaU, and (c) KSC.

parameter. All parameters are fixed during parameter tuning except for the one being tuned.

First, we traverse the number of superpixels S_I, whose value is chosen from {1, 3, 5, 10, 20, 30, 40, 50, 75, 100, 150, 200, 300} for the three datasets. Fig. 8 illustrates the OA of MDT-DSGCN as a function of the initial number of superpixels in ERS.

In the MDT-DSGCN, with the initial number of superpixels increasing in the ERS, on the Indian Pines and PaviaU, the OA first increases and then decreases. The reason is that many superpixels can lead to over-segmentation, causing superpixels to degenerate into ordinary pixels and not be fully used. Fewer superpixels may lead to under-segmentation, introducing samples from different homogeneous areas and resulting in more misclassifications. For KSC, the distribution of its surface feature samples is too scattered, with a total sample size accounting for only 1.7% of the spatial dimension, much smaller than PaviaU (20.62%) and Indian Pines (48.75%), resulting in a continuous decline in CA.

The number of neighborhood scales C also affects MDT-DSGCN, with Fig. 9 presenting the OA of MDT-DSGCN as a function of the neighborhood scales. Fig. 9 highlights that OA increases with the number of neighborhood scales because we establish multiple input graphs with different neighborhood scales to extensively exploit the diversified spectral-spatial correlations at multiple scales.

Next, for another branch network, HR-ACPCNN, we discuss two parameters related to spatial-spectral joint information: the spatial window size of each neighboring cube S_N in hybrid CNN, and the reduced spectral dimension D_P after PCA. For parameter D_P, different datasets have different spectral dimensions. The value range is {10, 20, 30, 40, 50, 75, 100, 110, 120, 130, 140, 175} for Indian Pines, {15, 20, 30, 40, 50, 75, 100} and {10, 20, 30, 40, 50, 75, 100, 110, 120, 130, 140, 150} for PaviaU and KSC. Figs. 10 and 11 show the OA of the HR-ACPCNN under two different parameters. In the HR-ACPCNN network, Fig. 10 reveals that as the spatial window S_N increases, the OA first increases and then decreases. Similarly, the impact of the parameter D_P on OA is the same as S_N (Fig. 11).

TABLE VII
EXPERIMENTAL HYPERSPECTRAL PARAMETERS SETTINGS IN HDECGCN

Dataset	S_I	C	S_N	D_P
Indian Pines	100	4	11*11	140
KSC	1	4	11*11	120
PaviaU	20	4	11*11	30

Finally, Table VII summarizes the four parameters in the HDECGCN framework and provides the hyperspectral parameter settings for subsequent comparative experiments.

D. Comparison With State-of-the-Art Method

To verify the advantages of HDECGCN in CA and runtime, we use Indian Pines, KSC, and PaviaU as the evaluation datasets. The specific details of the competitor algorithms are presented in Section IV-B. In order to compare and evaluate the CA of HDECGCN effectively, all algorithms are evaluated ten consecutive times, and the average OA, AA, Kappa coefficients and running time are obtained. Tables VIII–X report different algorithms' CA and running time on the three datasets.

In Tables VIII–X, the optimal values per-class accuracy, OA, AA, and Kappa coefficients for each of the ten classification algorithms are highlighted in red, and the longest-running time is presented in bold.

For the Indian Pines dataset, HDECGCN achieves optimal values of 99.14%, 99.28%, and 0.9904 for OA, AA, and Kappa coefficients, respectively. HDECGCN has a lower standard deviation in all three evaluation indicators than the other eight algorithms. Compared to the MSuperPCA, HDECGCN uses GPU devices and gradually aggregates low-level features to automatically obtain abstract high-level nonlinear features,

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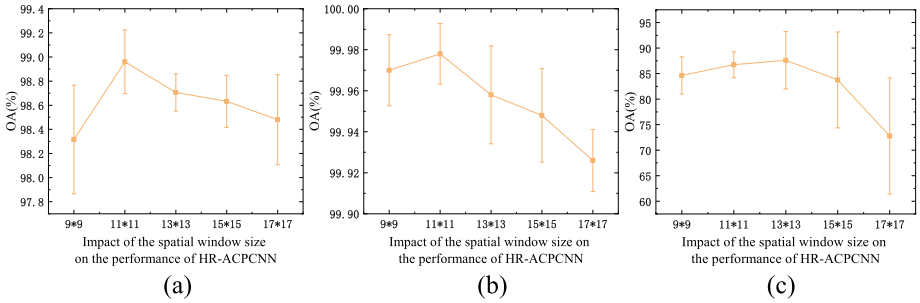


Fig. 10. Influence of the size of the spatial window S_N on the OA (%) of the three datasets (a) Indian Pines, (b) PaviaU, and (c) KSC.

TABLE VIII
COMPARISON RESULTS OBTAINED ON THE INDIAN PINES DATASET. WE RANDOMLY SELECT 10% OF THE SAMPLE SET TO CONSTRUCT THE TRAINING SAMPLES

Class No	MSuperPCA	SSRN	HybridSN	MCNN-CP	MDGCN	DBDA	CEGCN	AMGCFN	WFCG	OUR
1	97.73	**100**	96.55	99.19	86.10	88.72	95.34	98.02	99.50	**100**
2	96.81	97.93	**99.84**	96.81	95.59	98.33	98.82	98.74	99.22	97.33
3	95.39	95.72	97.63	99.47	96.20	94.19	98.97	98.94	98.79	**100**
4	94.71	**100**	**100**	95.56	95.45	98.47	98.86	98.74	97.84	99.15
5	98.89	98.85	95.43	98.48	95.43	98.31	99.65	98.50	98.70	**99.95**
6	99.01	**100**	99.39	99.33	97.64	99.19	99.63	99.47	99.86	99.74
7	**100**	**100**	91.30	99.09	86	84.36	99.60	94.38	99.09	**100**
8	**100**	**100**	**100**	**100**	99.67	99.32	99.88	99.95	99.95	**100**
9	95.45	**100**	**100**	63.75	93.33	84.39	65.56	**100**	90.00	**100**
10	96.68	94.58	96.66	98.16	93.26	94.94	97.63	97.56	98.62	**98.91**
11	96.52	96.20	98.65	99.20	97.75	97.41	**99.63**	99.53	99.06	99.39
12	94.97	**98.83**	96.34	97.85	94.74	95.15	98.81	98.68	97.07	97.24
13	96.53	**100**	95.48	98.54	99.03	99.70	99.67	99.66	99.78	99.57
14	**99.92**	99.56	99.56	99.86	99.67	98.73	99.90	99.76	99.32	99.28
15	97.84	**100**	**100**	98.15	98.99	98.10	97.48	99.82	98.21	99.88
16	97.89	98.46	98.46	90.40	94.76	92.75	97.70	96.09	**99.28**	97.47
OA(%)	97.26 ±0	97.48 ±0.23	98.33 ±0.10	98.52 ±0.11	96.83 ±0.43	97.16 ±1.84	99.07 ±0.19	99.07 ±0.27	98.96 ±0.26	**99.14 ±0.12**
AA(%)	97.40 ±0	98.76 ±1.77	97.83 ±2.43	95.87 ±1.58	95.21 ±0.80	95.13 ±6.73	96.70 ±2.02	98.62 ±0.62	98.39 ±2.16	**99.28 ±0.14**
Kappa*10 0	96.85 ±0	97.12 ±0.26	98.09 ±0.10	98.31 ±0.13	96.39 ±0.49	96.76 ±2.09	98.94 ±0.71	98.94 ±0.31	98.82 ±0.30	**99.04 ±0.17**
Time(s)	109.81	**330.14**	137.31	137.17	36.79	93	151.56	21.01	15.78	156.66

thus increasing OA by 1.88%. Compared to SSRN, HybridSN, and MCNN-CP based on convolutional network layer stacking, HDECGCN achieves complementarity between ordinary pixels and superpixels. Specifically, the MDT-DSGCN branch establishes dynamic multiscale input maps of different neighborhoods. The input map is a superpixel map containing average spectral features obtained through two consecutive superpixel divisions. This superpixel map stores local structural information of HSIs, and the MDT-DSGCN branch learns multiscale irregular features. Additionally, the convolutional kernel of the HR-ACPCNN performs convolution on small-scale regular regions. Compared to the MDT-DSGCN branch, which considers large-scale superpixel blocks as nodes in the graph, the HR-ACPCNN branch generates ordinary pixel-level features to supplement the problem of small domain classification errors caused by superpixel segmentation in the MDT-DSGCN branch and uses group convolution and residual nonlocal blocks to reduce overfitting and capture long-range dependencies. As a result, OA increases by 1.66%, 0.81%, and 0.62%, respectively.

For the KSC dataset, HDECGCN achieves optimal values of 99.43% and 99.26% on OA and AA, respectively, while the lower standard deviation ensured stable CA of the entire framework. Compared to the 10% Indian Pines training set, the OA increases by 0.29% on only 6% of the training set. The reason is that the total sample size of KSC accounts for a small proportion of pixels in the image spatial domain, only 1.7%. Therefore, it has a higher spatial resolution and is easier to achieve high CA with fewer training samples. Regarding CA, the HDECGCN framework improves OA by 2.23% compared to DBDA, the reason is that HDECGCN has the ability to simulate long-range spatial relationships in HSIs, which DBDA does not have. Our method achieves 100% CA in all eight feature types, demonstrating its superiority in CA.

For the PaviaU dataset, HDECGCN achieves 99.93% and 99.88% on OA and AA, with a standard deviation value of

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TABLE IX

COMPARISON RESULTS OBTAINED ON THE KSC DATASET. WE RANDOMLY SELECT 6% OF THE SAMPLE SET TO CONSTRUCT THE TRAINING SAMPLES

Class No	MSuperPCA	SSRN	HybridSN	MCNN-CP	MDGCN	DBDA	CEGCN	AMGCFN	WFCG	OUR
1	97.58	95.32	92.71	96.64	99.66	**100**	98.90	**100**	99.39	**100**
2	**100**	97.54	70.15	70.37	89.30	95.03	99.38	99.60	99.43	98.84
3	95.82	97.40	85.47	78.05	92.28	90.54	99.70	99.96	99.07	**100**
4	**97.61**	87.08	50.44	57.75	88.23	80.99	92.68	95.67	91.98	**97.47**
5	92.21	81.24	90.62	72.06	98.74	82.20	95.06	92.78	95.64	**99.48**
6	89.57	84.86	70.25	77.61	91.67	95.93	99.30	**99.95**	96.46	97.16
7	70.31	93.33	66.30	92.15	90.71	95.28	99.48	98.26	99.44	**100**
8	**100**	98.62	82.91	91.87	94.91	99.14	99.43	99.98	99.30	**100**
9	97.10	99.22	71.77	95.76	98.83	**100**	**100**	**100**	99.88	97.47
10	90.21	97.53	65.32	83.90	97.89	**100**	**100**	**100**	99.59	**100**
11	99.21	99.78	95.15	99.24	99.70	99.47	**100**	**100**	99.97	**100**
12	96.83	99.24	78.18	83.97	99.15	99.41	99.76	99.13	99.96	**100**
13	**100**	**100**	98.94	99.96	**100**	99.88	**100**	**100**	100	**100**
OA(%)	96.31 ±0	96.61 ±0.36	82.55 ±2.43	89.04 ±1.34	97.11 ±0.93	97.20 ±1.13	99.18 ±0.39	99.42 ±0.36	99.01 ±0.43	**99.43 ±0.45**
AA(%)	94.34 ±0	94.71 ±0.73	78.32 ±2.72	84.57 ±1.63	95.47 ±1.08	95.22 ±6.69	98.75 ±0.65	98.87 ±0.78	98.47 ±0.74	**99.26 ±0.51**
Kappa*10 0	95.89 ±0	96.22 ±0.40	80.53 ±2.71	87.79 ±1.49	96.77 ±1.03	96.88 ±0.52	99.08 ±0.44	99.26 ±0.40	98.90 ±0.48	**99.28 ±0.40**
Time(s)	116.89	243.67	61.11	48.33	17.78	43.89	**3314.40**	2903.12	**4466.33**	263.44

TABLE X

COMPARISON RESULTS OBTAINED ON THE PAVIAU DATASET. WE RANDOMLY SELECT 6% OF THE SAMPLE SET TO CONSTRUCT THE TRAINING SAMPLES

Class No	MSuperPCA	SSRN	HybridSN	MCNN-CP	MDGCN	DBDA	CEGCN	AMGCFN	WFCG	OUR
1	97.78	99.29	99.62	99.93	99.65	99.71	99.97	**99.97**	99.90	**99.99**
2	99.72	99.71	99.99	99.94	99.95	99.88	**100**	**100**	99.91	**100**
3	99.61	96.92	98.42	98.40	**100**	99.60	99.94	99.94	99.93	99.55
4	97.88	**99.89**	98.22	92.66	97.16	98.38	99.32	99.15	99.81	99.65
5	99.72	99.52	99.12	98.85	99.68	**100**	**100**	99.98	99.94	**100**
6	99.60	99.87	**100**	99.99	**100**	99.88	**100**	**100**	**100**	99.99
7	98.85	99.76	99.31	99.49	99.79	99.92	**100**	**100**	99.99	99.96
8	98.85	97.36	97.51	99.68	**99.81**	97.04	99.77	99.64	99.60	99.76
9	99.85	99.66	91.98	92.06	93.21	98.93	99.94	98.76	99.92	**99.98**
OA(%)	99.17 ±0	99.33 ±0.29	99.29 ±0.22	99.10 ±0.14	99.54 ±0.28	99.46 ±0.24	99.92 ±0.02	99.87 ±0.06	99.89 ±0.04	**99.93 ±0**
AA(%)	99.10 ±0	99.16 ±0.18	98.24 ±0.49	97.89 ±0.25	98.81 ±0.41	99.26 ±0.29	**99.88** ±0.02	99.72 ±0.13	99.89 ±0.05	**99.88 ±0.02**
Kappa*10 0	98.88 ±0	99.14 ±0.22	99.06 ±0.28	98.81 ±0.18	99.39 ±0.33	99.28 ±0.31	99.90 ±0.02	99.83 ±0.07	99.85 ±0.05	**99.91 ±0.01**
Time(s)	690.14	93.53	181.84	203.66	76.16	217.19	**1795.05**	767.34	**1616.49**	615.60

0 and 0.02%. Compared to MDGCN using a single GCN, the OA improves by 0.39% and the AA by 1.07% on a 6% training set because we propose a dual superpixel segmentation to obtain compact and uniform superpixel blocks. Next, we use the average standard deviation to calculate the weight coefficients of superpixels between different neighborhoods. Experiments show that this structure can improve the optimal number of scales (from 3 to more than 4) compared to MDGCN, optimizing the topological structure of the graph. Compared with the CEGCN, although the AA of HDECGCN is equal to those of the CEGCN, the running time of CEGCN is 1795 s, almost three times the running time of our framework. Our framework's OA on the KSC and Indian Pines are 0.25% and 0.07% higher than the CEGCN, maintaining a minimal standard deviation on Indian Pines, the reason is that HDECGCN focuses on optimizing classification results, using a decision fusion strategy instead of the encoder-decoder in CEGCN, which reduces the model's computation time while maintaining compatibility with the two subnetworks. Compared with the WFCG, the OA increases by 0.04%, the reason is that HDECGCN obtains the multiscale and high-level abstract features of HSI.

The running time of all algorithms is reported in the last row of Tables VIII–X, and the algorithms' running environment is presented in Section IV-B. The MSuperPCA is based on multiscale superpixel segmentation, which consumes too much

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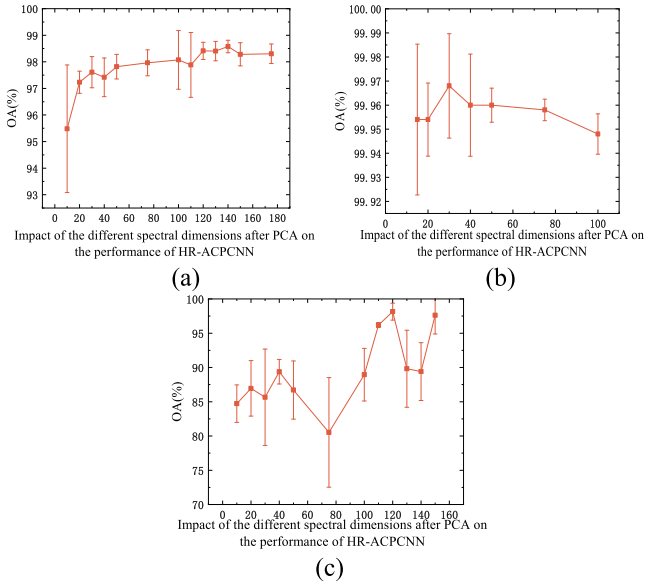


Fig. 11. Influence of the spectral dimensions after PCA on the OA (%) of the three datasets (a) Indian Pines, (b) PaviaU, and (c) KSC.

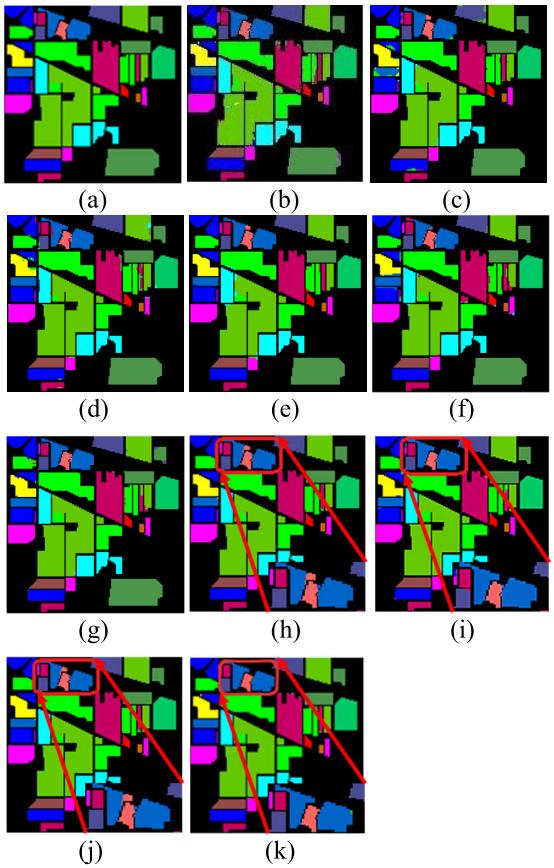


Fig. 12. Indian Pines (a) reference image, (b) MSuperPCA (OA = 97.26%), (c) SSRN (OA = 97.50%), (d) HybridSN (OA = 98.23%), (e) MCNN-CP (OA = 98.61%), (f) MDGCN (OA = 97.05%), (g) DBDA (OA = 98.22%), (h) CEGCN (OA = 99.07%), (i) AMGCFN (OA = 99.19%), (j) WFCG (OA = 99.20%), and (k) HDECGCN (OA = 99.28%).

time on PaviaU and has low CA. The SSRN requires more epochs to train the network, and the MCNN-CP is a structure obtained by modifying the HybridSN. Thus, the running times of the two algorithms are not significantly different. MDGCN uses superpixel blocks as input units for GCN, thereby reducing runtime and achieving the highest efficiency among all algorithms. The CEGCN consumes significant time on PaviaU and KSC. Our framework sets the epochs of the two branch networks to 50 and 4000, respectively, reducing the number of epochs and running time while maintaining high CA on the three datasets.

The classification maps obtained with the abovementioned three public HSI datasets for the HDECGCN framework and the nine comparison methods are presented in Figs. 12–14.

Figs. 12–14 illustrate the classification maps using MSuperPCA, SSRN, HybridSN, MCNN-CP, MDGCN, DBDA, CEGCN, AMGCFN, WFCG, and HDECGCN. Here, we only show the results when the number of training samples is set to 10%, 6%, and 6% of the total samples, respectively. These maps highlight that the MSuperPCA exhibits higher classification errors than the other methods, proving that deep networks can extract high-level nonlinear features of images. Besides, the quality of the CEGCN and HDECGCN classification maps is far better than that of the other methods for the three datasets. The MDGCN based on a single branch GCN is weaker than the SSRN, HybridSN, and MCNN-CP algorithms of a single branch CNN on the Indian Pines. For the KSC, the OA of MDGCN is higher than that of SSRN, HybridSN, and MCNN-CP of single-branch CNN. The reason is that the KSC has higher resolution and is more suitable for GCNs to establish middle- and long-range spatial relations through its graph structure. The DBDA uses a double-branch dual-attention mechanism network to capture many spectral-spatial features in HSI. On the Indian Pines and PaviaU, the OA of DBDA are 0.72% and 0.53% higher than SSRN's, respectively. Like our framework, the adaptive attention mechanism and the Mish capture spatial-spectral joint features and accelerate the convergence process. In contrast, fusing the CNNs and GCNs in the CEGCN achieves a stable and high CA. However, as the number of training samples increases, the time-consuming in the encoding and decoding stages gradually increase. When facing PaviaU and KSC that require classification, the CEGCN is unsuitable. The CA of AMGCFN on the three datasets is better than that of single-branch CNNs and GCNs networks and lower than that of the HDECGCN, the reason is that we leverage spectral information in both subnetworks to achieve a balance between attention mechanisms and spectral information extraction. In order to maintain high CA and reduce runtime, our framework uses multiscale dynamic tuning parameters, such as GCN fusion hybrid CNN, while reducing the number of epochs used in dual network branches. The randomly generated classification maps on three datasets achieve the highest OA of 99.28%, 99.52%, and 99.93%, respectively. Next, to demonstrate the robustness of the framework in classification performance, we verify the superiority of the OA of the HDECGCN framework under different training samples (Fig. 15), especially proving the superiority of OA on small training samples.

Fig. 15 further shows the performance of all competitor methods under increasing training samples. Specifically, the ratios of training samples are set from 0.5% to 30% per class for the Indian Pines and the PaviaU, and the ratio of

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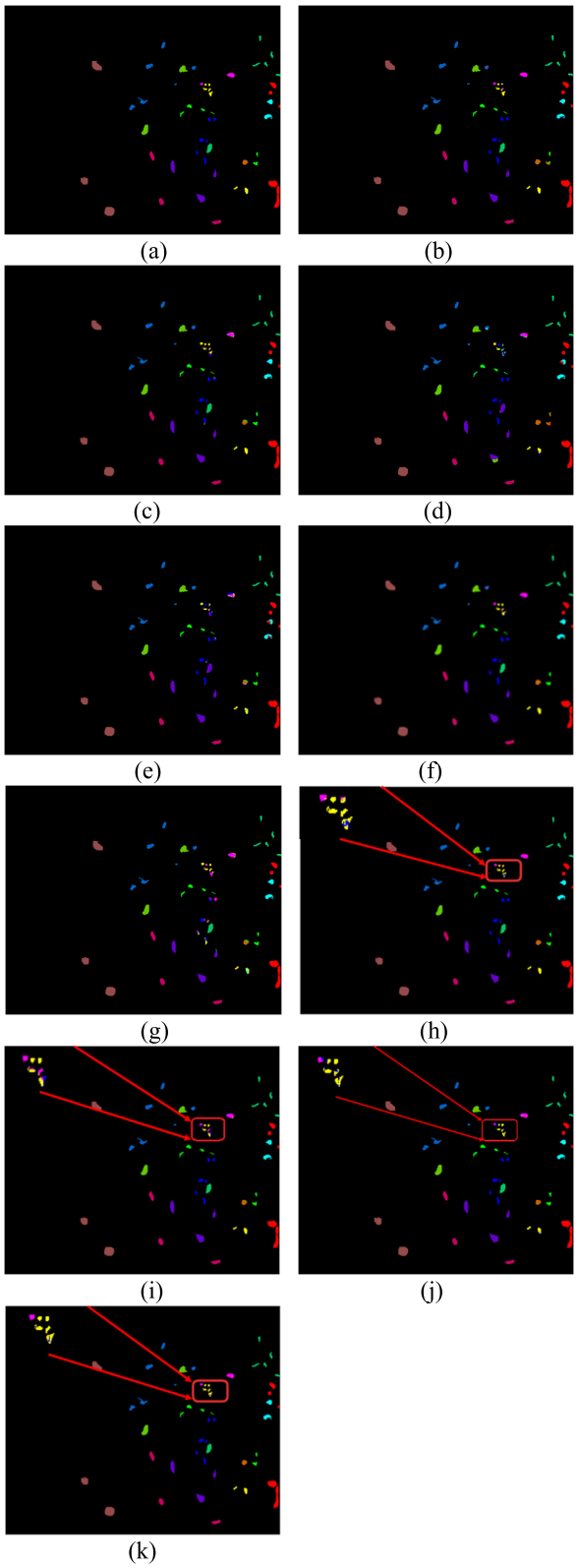


Fig. 13. KSC (a) reference image, (b) MSuperPCA (OA = 96.31%), (c) SSRN (OA = 96.67%), (d) HybridSN (OA = 87.84%), (e) MCNN-CP (OA = 89.52%), (f) MDGCN (OA = 97.73%), (g) DBDA (OA = 93.90%), (h) CEGCN (OA = 99.08%), (i) AMGCFN (OA = 98.62%), (j) WFCG (OA = 98.85%), and (k) HDECGCN (OA = 99.52%).

training samples for the KSC is set from 3% to 20% per class. HDECGCN achieves the best performance for Indian Pines and KSC. Although the CEGCN, WFCG, and AMGCFN

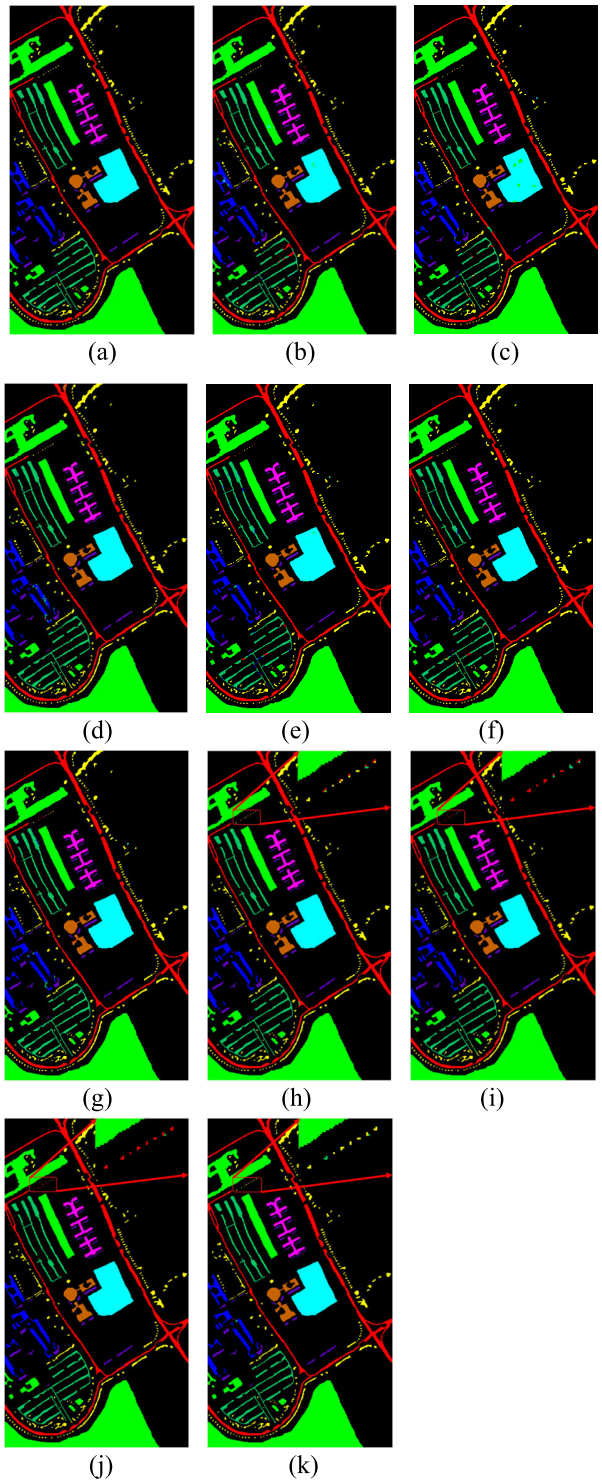


Fig. 14. PaviaU (a) reference image, (b) MSuperPCA (OA = 99.17%), (c) SSRN (OA = 99.03%), (d) HybridSN (OA = 99.26%), (e) MCNN-CP (OA = 99.11%), (f) MDGCN (OA = 99.47%), (g) DBDA (OA = 99.56%), (h) CEGCN (OA = 99.91%), (i) AMGCFN (OA = 99.77%), (j) WFCG (OA = 99.85%), and (k) HDECGCN (OA = 99.93%).

achieve higher OA on the PaviaU than HDECGCN when the training size is small, However, the running time of the three algorithms is too high. For example, we found through experiments that the three algorithms achieved 2542.33 (CEGCN), 1530.66 (WFCG), and 960.55 s (AMGCFN) on 1% of the training samples, respectively, HDECGCN took 524.91 s. In summary, HDECGCN excels in the classification of HSIs

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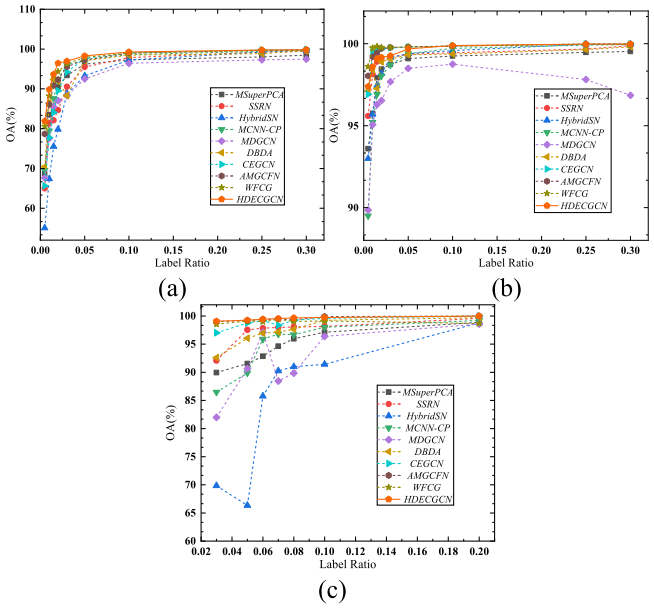


Fig. 15. Classification results (%) using different models on different training size amounts (a) Indian Pines, (b) PaviaU, and (c) KSC.

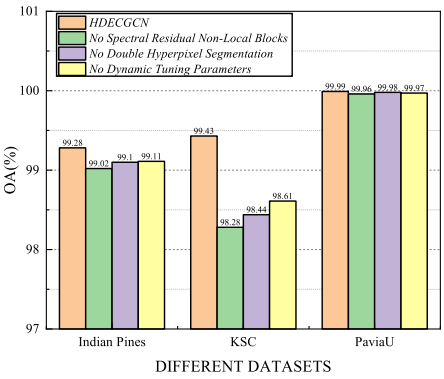


Fig. 16. Validation in HDECGCN of each newly added module.

on small-sample datasets, maintaining a balance between CA and running time.

V. Discussion

A. Ablation Study

To show the performance gain from newly added modules, we conduct an ablation study of newly added modules that may impact downstream classification task performance, especially on the three datasets: Indian Pines, PaviaU, and KSC. These effective gain modules encompass spectral stacking residual nonlocal blocks, double hyperpixel segmentation, and dynamic tuning parameters. The ablation experiment results of each module on three datasets are shown in Fig. 16.

It can be observed from Fig. 16 that we sequentially compare the HDECGCN framework with these modules: On the basis of HDECGCN, only delete the spectral stacking residual nonlocal block. Second, only use the SLIC segmentation algorithm, retaining the other two innovations. Finally, in the HDECGCN, only the dynamic tuning parameters are not used.

According to Fig. 16, compared with the other two modules, the spectral residual nonlocal block is the module that affects the OA of HDECGCN the most. This also proves that the attention mechanism is all our need. On Indian Pines and PaviaU, the gains of dual superpixel segmentation and dynamic tuning parameters on HDECGCN are almost the same. On KSC, the gain of dual superpixel segmentation is greater than that of dynamic tuning parameters, the reason is that the low label ratio (1.7%) of the KSC dataset requires optimization of the nodes in the graph. In summary, the newly added modules have a positive impact on the final classification results and can effectively improve the OA.

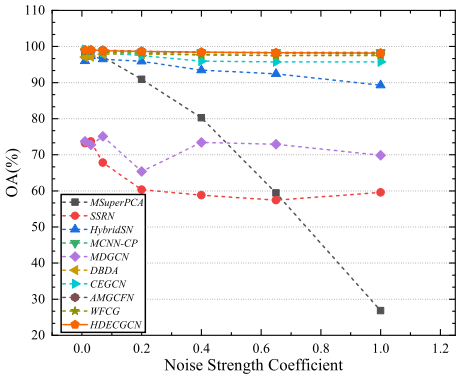


Fig. 17. OA(%) of the ten HSI classification algorithms under different additive Gaussian noise intensities (on Indian Pines, execute ten times and calculate the average value).

B. Robustness Study

When meets the degraded HSIs, to quantitatively validate the robustness of HDECGCN [50]. Specifically, the study is divided into two parts. Consider that in the observation model $\boldsymbol{H}_o = \boldsymbol{H}_c + \boldsymbol{N}_S$, $\boldsymbol{H}_c$ is a clean image, $\boldsymbol{N}_S$ is zero-mean, additive, Gaussian, pixelwise independent with spectral covariance [51]. We set a noise intensity coefficient ω. As can be clearly seen from Fig. 17, HDECGCN is more robust and effective against noises with different strengths, specifically, when the noise intensity coefficients are 0.01, 0.03, and 0.4, HDECGCN performs better and more robust than other comparative algorithms. When the intensity coefficient is 0.07 or 0.2, HDECGCN is only weaker than AMGCFN, when the intensity coefficient is greater than 0.65, it is only weaker than MCNN-CP but higher than CEGCN, WFCG, and AMGCFN. The OA value has been consistently greater than 98%. Besides, we experimentally discuss another case of all spectral bands' fog coverage as input [52]. More specifically, add the same-value white noise uniformly for every other pixel on the band, and then run the Gaussian filter on the whole band twice, the kernel size is 3×3, and no padding. Table XI shows the average OA of the ten HSI classification algorithms under the fog, HDECGCN achieved an OA of 98.92%, just lower than CEGCN and WFCG. It can be concluded that HDECGCN can adapt to complex Gaussian noise or foggy environments.

C. Computational Complexity Analysis

For an HSI with spectral band length N, due to the fact that the HDECGCN includes two sub-networks: HR-ACPCNN and MDT-DSGCN. The SBNL yields the complexity of $O(N/2)$,

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TABLE XI
OA (%) OF THE TEN HSI CLASSIFICATION ALGORITHMS UNDER THE COVER OF FOG (ON INDIAN PINES, EXECUTE TEN TIMES AND CALCULATE THE AVERAGE VALUE)

	MSuperPCA	SSRN	HybridSN	MCNN-CP	MDGCN	DBDA	CEGCN	AMGCFN	WFCG	OUR
OA(%)	97.90 ±0	98.78 ±0.07	96.79 ±2.21	98.71 ±0.15	69.51 ±2.17	98.27 ±0.61	**99.13** ±0.13	98.92 ±0.37	98.83 ±0.41	98.92 ±0.04
Time(s)	46.81	330.82	139.75	483.94	35.92	136.64	145.24	28.16	**15.85**	187.18

TABLE XII
DIFFERENCE IN CA OF GROUND FEATURES WITH COMPLEMENTARY PHENOMENA ON THE INDIAN PINES (10% TRAINING SET)

Class Num	HR-ACPCNN	MDT-DSGCN	Differences in CA(%)
7(Grass-pasture)	√		16%
9(Oats)		√	62.5%
12(Soybean-clean)	√		11.87%

TABLE XIII
DIFFERENCE IN CA OF GROUND FEATURES WITH COMPLEMENTARY PHENOMENA ON THE KSC (6% TRAINING SET)

Class Num	HR-ACPCNN	MDT-DSGCN	Differences in CA(%)
4(Cabbage plam/oak)		√	53.53%
5(Slash pine)		√	12.51%
6(Oak/broadleaf hammock)		√	13.84%
8(Graminoid marsh)	√		16.87%

and the complexity of residual operations belongs to the constant level $O(1)$, so the computational complexity of the HR-ACPCNN is mainly caused by 3D-CNN, yielding an overall $O(\sum_{l=1}^{3} M_l^3 \times K_l^3 \times C_{\text{in}} \times C_{\text{out}})$, M_l, K_l, and C_{in} denote the length of the output feature map and convolutional kernel, and the number of output channels in the previous layer, C_{out} is the number of output channels. The MDT-DSGCN yields the complexity of $O(N \times S_{\text{SLIC}} + L \times |\boldsymbol{E}|)$, S_{SLIC} is the number of superpixels of the SLIC, and the edge $\boldsymbol{E}$ represents the relationships between pixels. To sum up, the computational complexity of the HDECGCN is $O(N \times S_{\text{SLIC}} + L \times |\boldsymbol{E}| + \sum_{l=1}^{3} M_l^3 \times K_l^3 \times C_{\text{in}} \times C_{\text{out}})$.

D. Interesting Points in the HDECGCN Framework

We reveal some interesting points about the HDECGCN framework by conducting several comparison experiments and parameter discussions on three HSIs as follows.

1) An HSI classification network architecture based on the fusion of CNNs and GCNs generates complementary spectral-spatial features at the pixel and superpixel levels, respectively. Specifically, we conduct experiments on the two branch networks, provide the surface features with complementary phenomena on the two branches, and define surface features with a CA difference of more than 10% due to complementary phenomena on the Indian Pines and KSC. The relevant results are presented in Tables XII and XIII, √ which means that the branch network has a higher CA.

Tables XII and XIII suggest that the complementary phenomena observed on the Indian Pines are grass structure, mowed oats, and soybean cleaning. The species on the KSC are cabbage palm/oak, slash pine, oak/broadleaf hammock, and graminoid marsh. Combining Tables III–VI, we find that the total number of samples for these features is relatively small. The MDT-DSGCN is more friendly for classifying small sample sizes of features because small samples of local features often have a more dispersed distribution, and the GCNs can capture the mid to long-distance spatial relationships between pixels. So, HDECGCN is more suitable for HSI classification tasks when facing sparse samples.

2) When facing datasets with PaviaU and KSC, the MSuperPCA and CEGCN have the longest running time compared to other algorithms. The reason is that when facing large training samples, even though the time complexity of the ERS algorithm is very low, the multiscale operation of MSuperPCA makes the calculation in ERS segmentation more complex, increasing the final running time. The CEGCN uses graph encoders and decoders to propagate features between image pixels and graph nodes, solving the incompatibility between CNNs and GCNs. As the number of samples increases, continuous encoding and decoding increases the complexity of matrix operations. In this case, our framework no longer performs multiscale segmentation on superpixels but applies it to graph construction. We perform decision fusion strategies on the classification maps generated at the pixel-superpixel level to replace the encoding-decoding operation. This strategy maintains high CA when facing large training samples and reduces the algorithm's runtime.

3) In Section IV-C, regarding the setting of initial pixel for ERS segmentation, for datasets with excessively dispersed distribution of surface feature samples (KSC), the OA gradually decreases as the initial pixel value increases because ERS segmentation produces superpixel blocks with only backgrounds that are invalid for the final classification results. As for the neighborhood scales C, our framework achieves a threshold C that increases by 1 compared to the MDGCN on three datasets. This proves that dual superpixel segmentation can mine more spatial-spectral joint information than the MDGCN. About the hyperparameters in the MDT-DSGCN, when facing high spatial size images (PaviaU and KSC), the setting of epoch values is higher than that of low spatial size image (Indian Pines) because it is difficult to capture long-distance relationships in high spatial size images.

We can conclude from the above that the parameter setting process should consider the size of HSI and the size of the included ground object samples. However, most deep learning

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based on HSI classification algorithms have fixed parameters, and the HDECGCN has changed this standard approach.

VI. Conclusion

This article introduces a novel heterogeneous framework, the multiscale dynamic tuning parameter dual superpixel segmentation GCNs, and fuses the enhanced hybrid 2-D–3-D CNNs for HSI classification (HDECGCN) for applications such as classifying precision agricultural features.

The HDECGCN framework consists of two branches: HR-ACPCNN and MDT-DSGCN. We introduce a three-channel stack to capture spectral features with discriminant information and a residual nonlocal block at the end of the hybrid 3-D–2-D groupable CNNs. The innovation of the MDT-DSGCN is based on the MDGCN, which uses SLIC to obtain the superpixels, but SLIC cannot fully obtain spatial information. In order to further utilize spatial information and maintain low computational time, we use the dual superpixel segmentation to process HSI and generate uniformly sized superpixel blocks for the entire region, allowing GCNs at each scale to capture better spatial information. Next, because the segmented superpixel blocks contain spectral information on different weights, we will use the average standard deviation to obtain the tuning parameters dynamically. Finally, we use the decision fusion algorithm instead of terminal fusion to achieve complementary performance of the framework. Experiments show that HDECGCN can achieve better results than the SOTA methods.

Although HDECGCN has achieved certain results in HSI classification tasks, there is still room for improvement as follows.

1) To establish a lightweight framework to reduce computational complexity, one proposal is to decrease the number of superpixel blocks while maintaining their compactness and uniformity, the idea is to automatically generate optimal pixel blocks based on the image. This would correspondingly decrease the computational complexity of the MDT-DSGCN sub-network.
2) Although the integration of parallel subnetworks can achieve the complementarity of pixels and super-pixels, determining the weights of the two subnetworks in the integration is the primary challenge for HDECGCN. In the future, these two sub-networks can be connected in series, and the weights of the two subnetworks can be determined based on different locations within the GCN.
3) The acquisition cost of labeled pixels is relatively high. When the number of labeled pixels in an image is limited, it can prevent HDECGCN from effective learning, thereby limiting the classification performance. In the future, self-supervision theories can be employed to explore the abundant supervisory information within HSIs, mitigating the decline in classification performance resulting from insufficient label information.
4) HDECGCN demonstrated high CA and reduced runtime on PaviaU and KSC. In the future, we will further explore the effectiveness of HDECGCN on the high spatial size of HSIs to verify the effectiveness of HDECGCN in the face of large-scale datasets.

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